

Data Handling And Visualization

LAB EXPERIMENTS (26-30)

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Set 26:

Student Mini Data Analysis

Dataset: Student_Mini_Data.csv

1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

CODE:

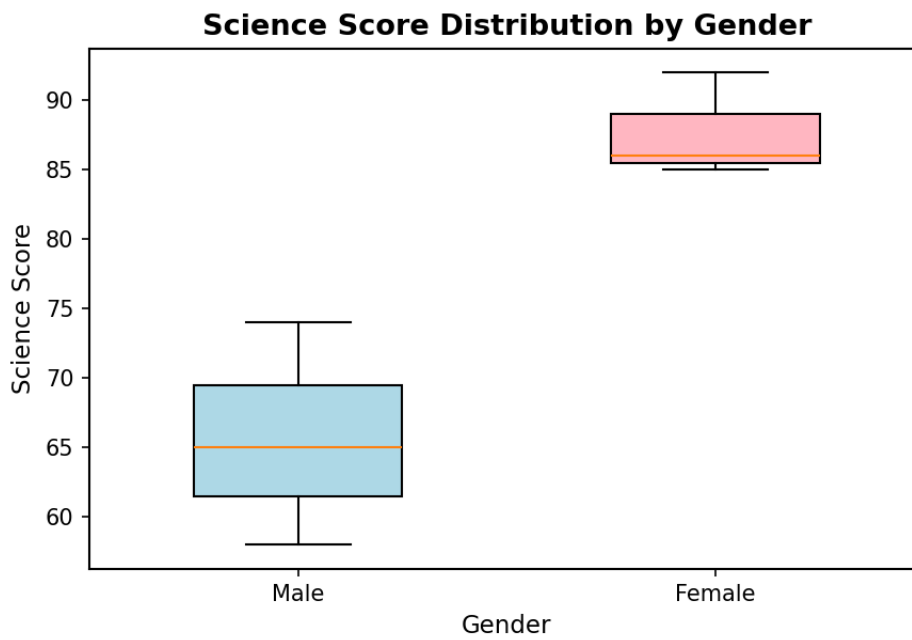
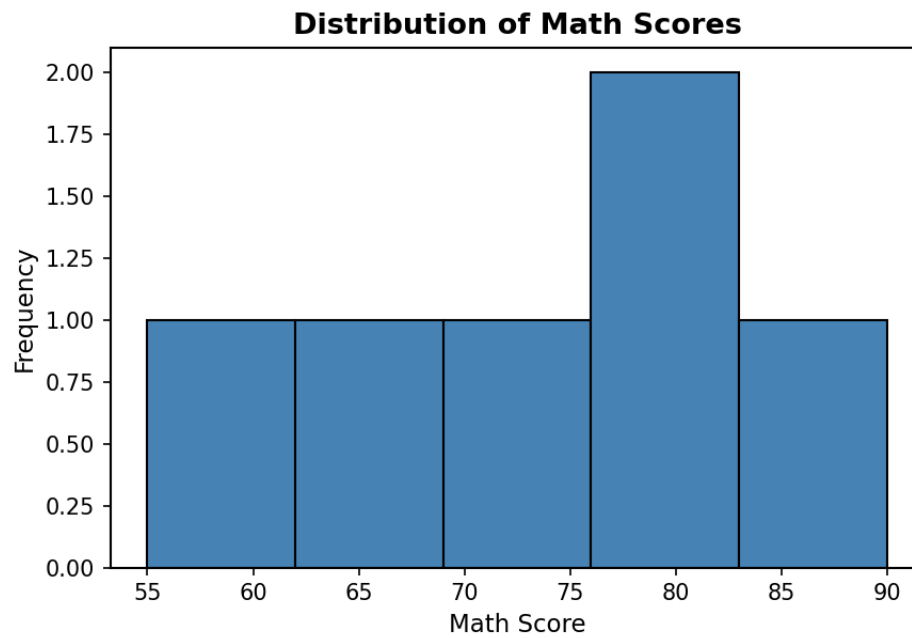
```
# Import dataset
student_data <- read.csv("Student_Mini_Data.csv")

# Histogram of Math_Score
hist(student_data$Math_Score,
      col = "steelblue",
      xlab = "Math Score",
      ylab = "Frequency",
      main = "Distribution of Math Scores")

# Boxplot of Science_Score by Gender

boxplot(Science_Score ~ Gender, data = student_data,
        col = c("lightpink", "lightblue"),
        xlab = "Gender",
        ylab = "Science Score",
        main = "Science Score Distribution by Gender")
```

OUTPUT:



Interpretation: Math scores are spread across the 55-90 range with no strong clustering, indicating varied performance among students. Female students show a visibly higher median Science score than male students, along with less spread, suggesting more consistent performance.

2. Create a scatter plot using R programming between Study_Time and Quiz_Score... (here: Study_Hours and Math_Score). Represent Gender using different colors. Add a regression line. Explain the relationship between study time and performance.

CODE:

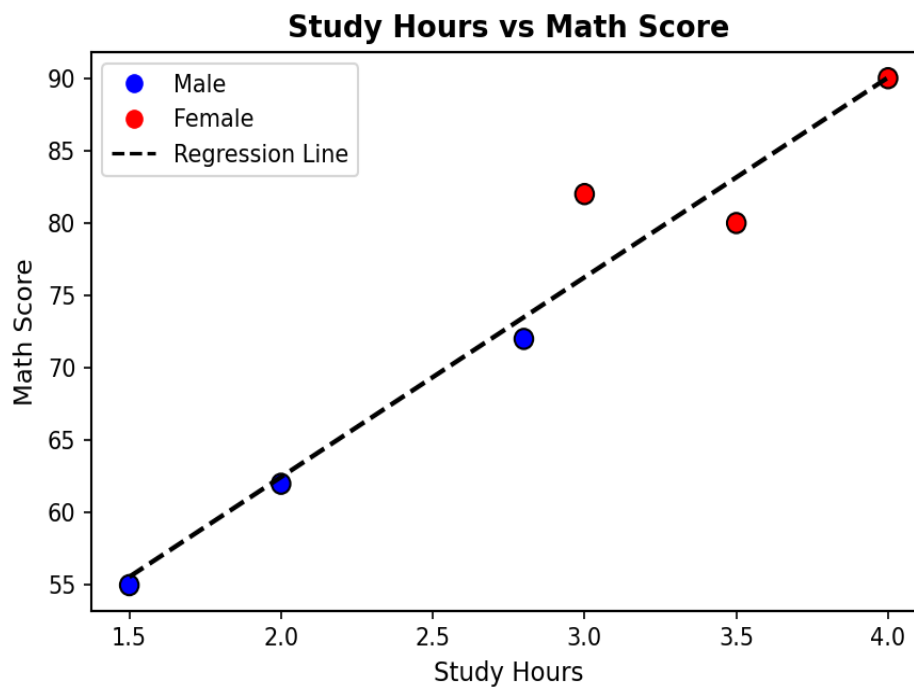
```
# Scatter plot of Study_Hours vs Math_Score
colors <- ifelse(student_data$Gender == "Male", "blue", "red")

plot(student_data$Study_Hours, student_data$Math_Score,
     col = colors, pch = 19, cex = 1.5,
     xlab = "Study Hours",
     ylab = "Math Score",
     main = "Study Hours vs Math Score")

# Regression line
model <- lm(Math_Score ~ Study_Hours, data = student_data)
abline(model, col = "black", lwd = 2, lty = 2)

legend("topleft",
     legend = c("Male", "Female", "Regression Line"),
     col = c("blue", "red", "black"),
     pch = c(19, 19, NA), lty = c(NA, NA, 2))
```

OUTPUT:



Interpretation: There is a clear positive relationship between study hours and Math scores - students who studied longer generally scored higher, as shown by the upward-sloping regression line. The pattern holds for both genders.

3. Using R programming, convert Login_Date (Exam_Date) into Date format. Calculate average quiz score (Math score) per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

CODE:

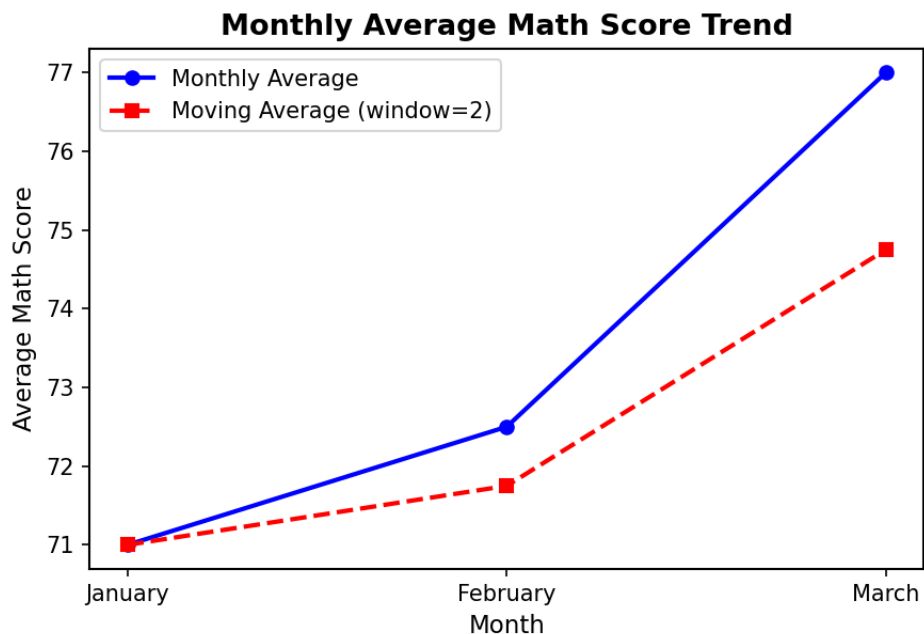
```
# Convert Exam_Date to Date format
student_data$Exam_Date <- as.Date(student_data$Exam_Date, format = "%Y-%m-%d")
student_data$Month <- format(student_data$Exam_Date, "%B")

# Average Math score per month
monthly_avg <- aggregate(Math_Score ~ Month, data = student_data, FUN = mean)

# Moving average smoothing (window = 2)
library(zoo)
monthly_avg$MA <- rollmean(monthly_avg$Math_Score, k = 2, fill = NA, align = 'right')

# Line chart
plot(monthly_avg$Math_Score, type = 'o', col = 'blue', lwd = 2,
      xaxt = "n", xlab = "Month", ylab = "Average Math Score",
      main = "Monthly Average Math Score Trend")
axis(1, at = 1:nrow(monthly_avg), labels = monthly_avg$Month)
lines(monthly_avg$MA, type = 'o', col = 'red', lwd = 2, lty = 2)
legend("topleft", legend = c("Monthly Average", "Moving Average"),
      col = c("blue", "red"), lty = c(1, 2), lwd = 2)
```

OUTPUT:



Interpretation: The average Math score rises steadily from January to March, indicating an overall improving trend across the recorded exam dates. The moving average line confirms this upward trajectory while smoothing out short-term fluctuation.

Set 27:

Patient Health Risk Analysis

Dataset: Patient Health Risk

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

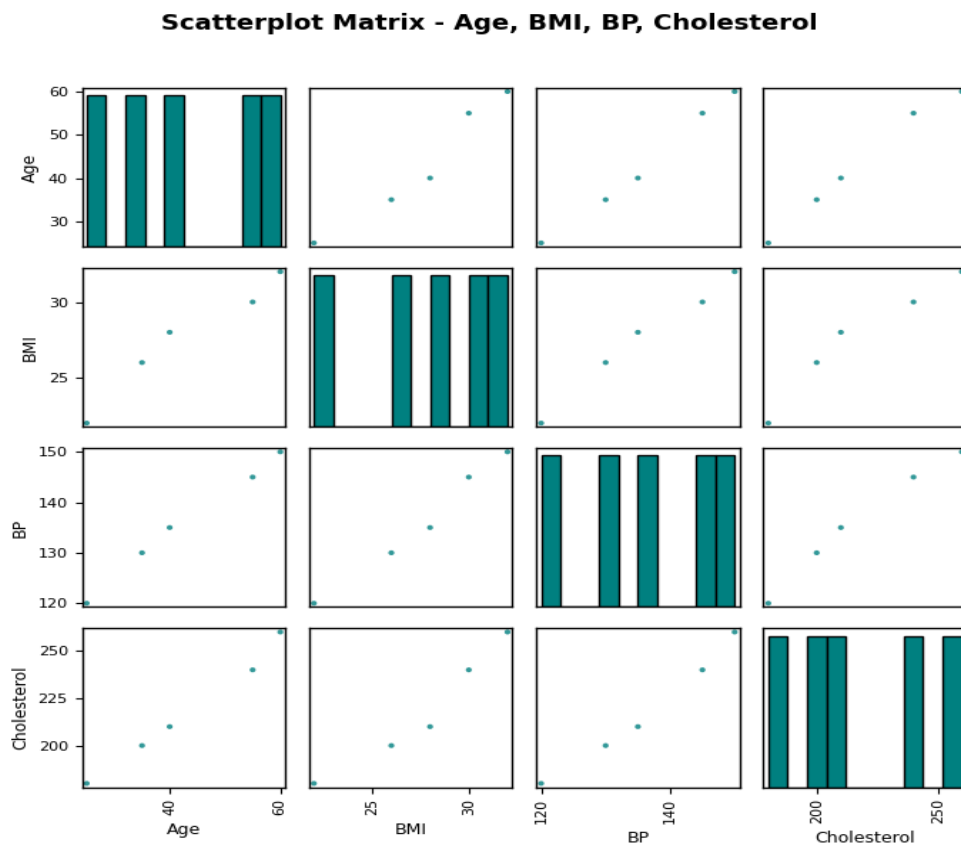
CODE:

```
# Dataset
patient_id <- c("P1","P2","P3","P4","P5")
age <- c(25, 40, 55, 35, 60)
bmi <- c(22, 28, 30, 26, 32)
bp <- c(120, 135, 145, 130, 150)
cholesterol <- c(180, 210, 240, 200, 260)

health_data <- data.frame(Age = age, BMI = bmi, BP = bp, Cholesterol = cholesterol)

# Scatterplot Matrix
pairs(health_data,
      col = "darkcyan", pch = 19,
      main = "Scatterplot Matrix - Age, BMI, BP, Cholesterol")
```

OUTPUT:



Interpretation: All four variables (Age, BMI, BP, and Cholesterol) move together in a strongly positive, near-linear pattern, so patients with a higher age consistently show a higher BMI, blood pressure, and cholesterol - a classic clustering of cardiovascular risk factors.

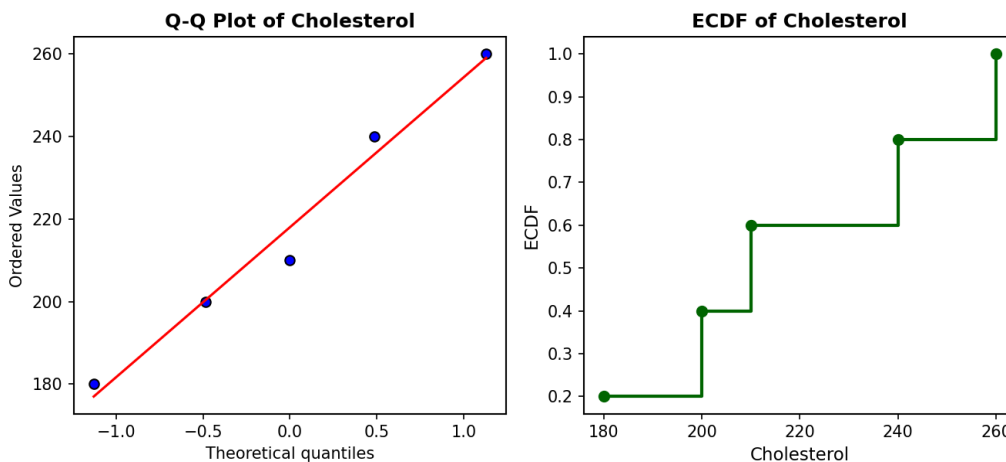
2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

CODE:

```
# Q-Q Plot for Cholesterol
qqnorm(cholesterol, main = "Q-Q Plot of Cholesterol", col = "blue", pch = 19)
qqline(cholesterol, col = "red", lwd = 2)

# ECDF for Cholesterol
plot(ecdf(cholesterol),
     col = "darkgreen", lwd = 2,
     xlab = "Cholesterol", ylab = "ECDF",
     main = "ECDF of Cholesterol")
```

OUTPUT:



Interpretation: The points on the Q-Q plot lie close to the reference line, suggesting Cholesterol values are approximately normally distributed with only mild deviation, and no strong skewness is evident. The ECDF confirms a smooth, evenly spaced increase across the observed range with no extreme outliers.

3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?

CODE:

```
# Average values of each indicator
averages <- c(Age = mean(age), BMI = mean(bmi),
              BP = mean(bp), Cholesterol = mean(cholesterol))

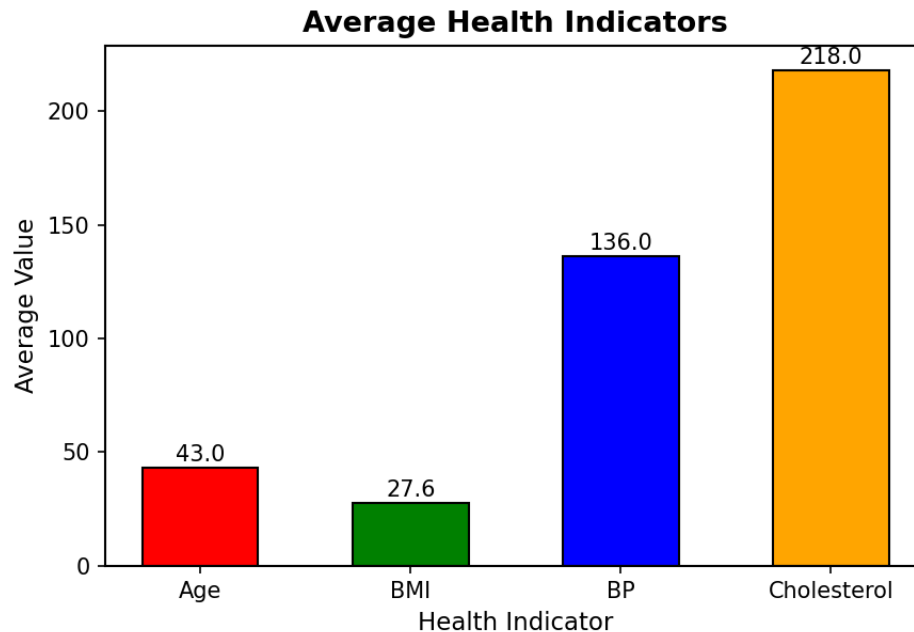
# Bar Chart
bar_positions <- barplot(averages,
                        col = c("red", "green", "blue", "orange"),
```

```

xlab = "Health Indicator",
ylab = "Average Value",
main = "Average Health Indicators")
text(bar_positions, averages + 3, labels = round(averages, 1))

```

OUTPUT:



Interpretation: The average BP (~136) and Cholesterol (~218) values fall above commonly cited healthy thresholds, while the average BMI (~27.6) sits in the overweight range - together these averages point to an elevated overall cardiovascular risk profile for this patient group.

Set 28:

Vehicle Performance Analysis

Dataset: Vehicle Performance

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

CODE:

```

library(vioplot)

# Dataset
vehicle_id <- c("V1", "V2", "V3", "V4", "V5")
engine_size <- c(1.5, 2.0, 3.0, 2.5, 1.8)
horsepower <- c(110, 150, 250, 200, 130)
fuel_efficiency <- c(18, 15, 12, 14, 17)

```

```

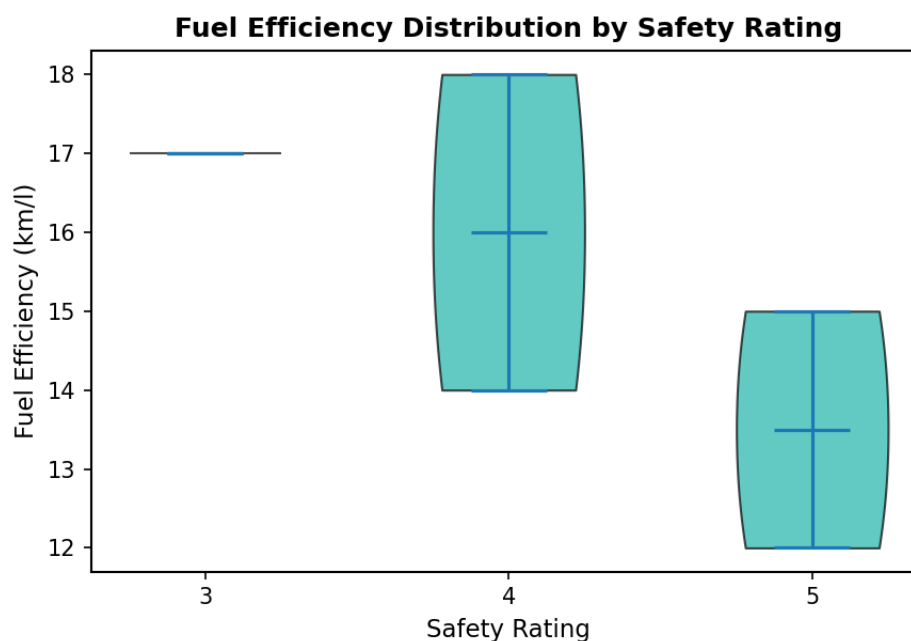
top_speed <- c(180, 200, 250, 220, 190)
safety_rating <- c(4, 5, 5, 4, 3)

vehicle_data <- data.frame(fuel_efficiency, safety_rating)

# Violin Plot
vioplot(fuel_efficiency ~ safety_rating, data = vehicle_data,
        col = "lightseagreen",
        xlab = "Safety Rating",
        ylab = "Fuel Efficiency (km/l)",
        main = "Fuel Efficiency Distribution by Safety Rating")

```

OUTPUT:



Interpretation: Vehicles with a lower safety rating (3) tend to have higher fuel efficiency, while the highest safety-rated vehicles (5) cluster around lower efficiency values, hinting at a trade-off where heavier, safer vehicles consume more fuel per kilometre.

2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

CODE:

```

# Scatter plot: Horsepower vs Top_Speed, colored by Engine_Size
col_ramp <- colorRampPalette(c("blue","green","yellow"))(100)
col_index <- as.numeric(cut(engine_size, breaks = 100))

plot(horsepower, top_speed,
     col = col_ramp[col_index], pch = 19, cex = 2,
     xlab = "Horsepower",

```

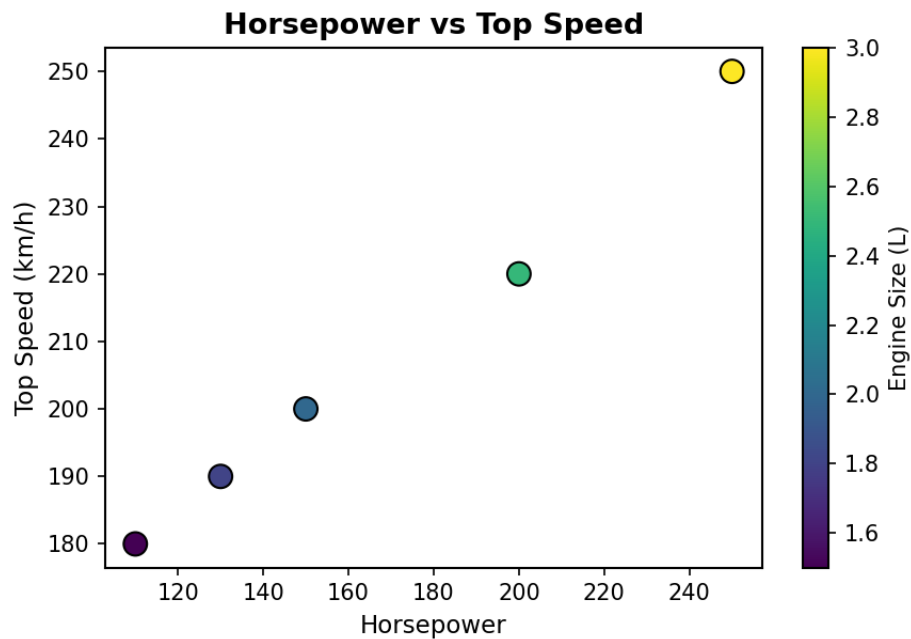
```

ylab = "Top Speed (km/h)",
main = "Horsepower vs Top Speed")

legend("topleft", legend = round(range(engine_size), 1),
      fill = col_ramp[c(1, 100)], title = "Engine Size (L)")

```

OUTPUT:



Interpretation: Horsepower and Top Speed show a strong positive correlation - as horsepower increases, top speed rises almost linearly. Larger engine sizes (shown in warmer colors) are associated with both higher horsepower and higher top speed, confirming engine size as a key performance driver.

3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

CODE:

```

library(reshape2)
library(ggplot2)

vehicle_data <- data.frame(engine_size, horsepower, fuel_efficiency,
                           top_speed, safety_rating)

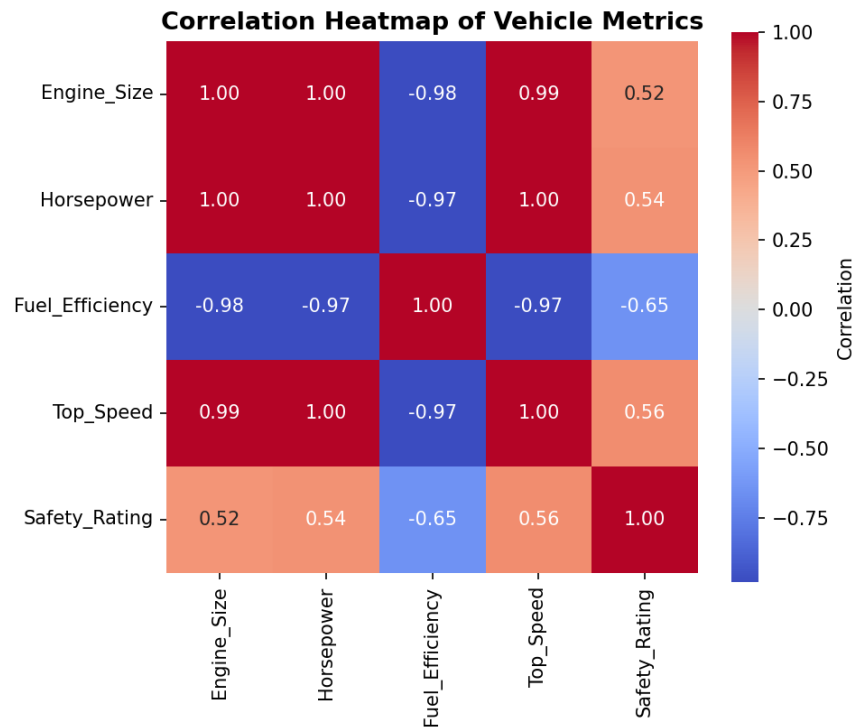
# Correlation matrix
cor_matrix <- cor(vehicle_data)
melted_cor <- melt(cor_matrix)

# Heatmap
ggplot(melted_cor, aes(Var1, Var2, fill = value)) +
  geom_tile() +

```

```
geom_text(aes(label = round(value, 2))) +
scale_fill_gradient2(low = "blue", high = "red", mid = "white",
                     midpoint = 0, limit = c(-1,1)) +
labs(title = "Correlation Heatmap of Vehicle Metrics") +
theme_minimal()
```

OUTPUT:



Interpretation: Top_Speed is most strongly and positively correlated with Horsepower and Engine_Size, and negatively correlated with Fuel_Efficiency, indicating that more powerful, larger-engine vehicles reach higher speeds but at the cost of efficiency.

Set 29:

Student Academic Performance

Dataset: Student Academic Performance

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

CODE:

```
# Dataset
student_id <- c("S1","S2","S3","S4","S5")
age <- c(19, 21, 20, 22, 23)
study_hours <- c(12, 8, 15, 10, 7)
attendance <- c(90, 70, 95, 85, 60)
```

```

test_score <- c(85, 70, 92, 80, 65)
participation_score <- c(8, 7, 9, 8, 6)

# Stacked Area Chart
x <- 1:5

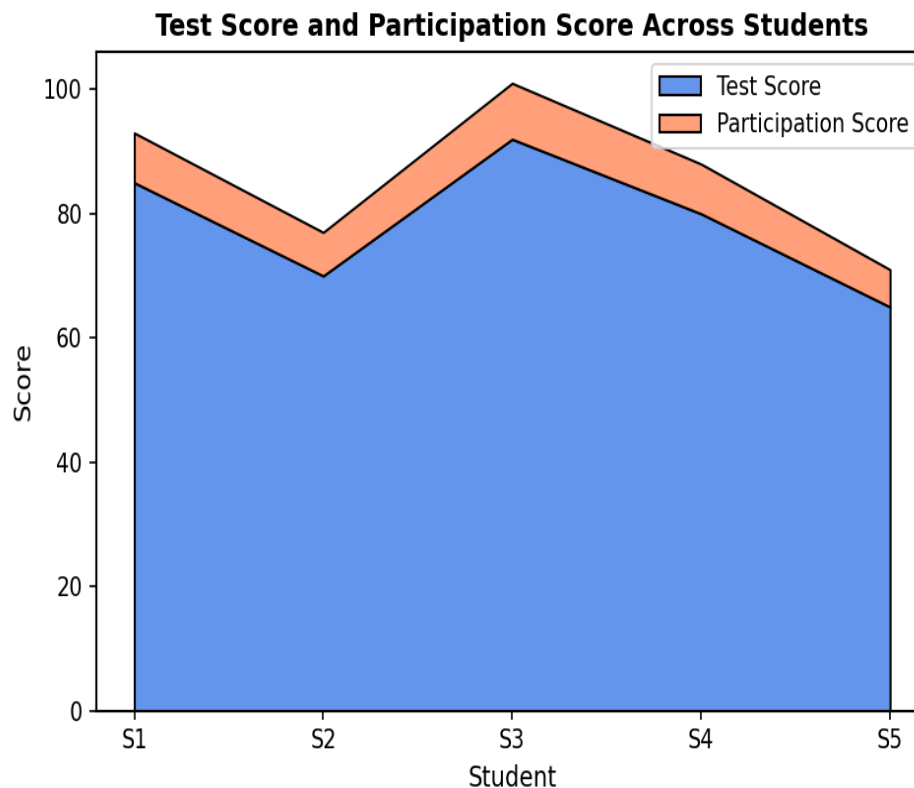
plot(x, test_score + participation_score, type = "n",
     xaxt = "n", xlab = "Student", ylab = "Score",
     main = "Test Score and Participation Score Across Students")
axis(1, at = x, labels = student_id)

polygon(c(x, rev(x)), c(rep(0,5), rev(test_score)),
       col = "cornflowerblue", border = "black")
polygon(c(x, rev(x)), c(test_score, rev(test_score + participation_score)),
       col = "lightsalmon", border = "black")

legend("topright", legend = c("Test Score", "Participation Score"),
      fill = c("cornflowerblue", "lightsalmon"))

```

OUTPUT:



Interpretation: Test scores dominate the combined total for every student and track participation scores closely - students with the highest test scores (S3) also show the highest participation, suggesting engagement and academic performance move together.

2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

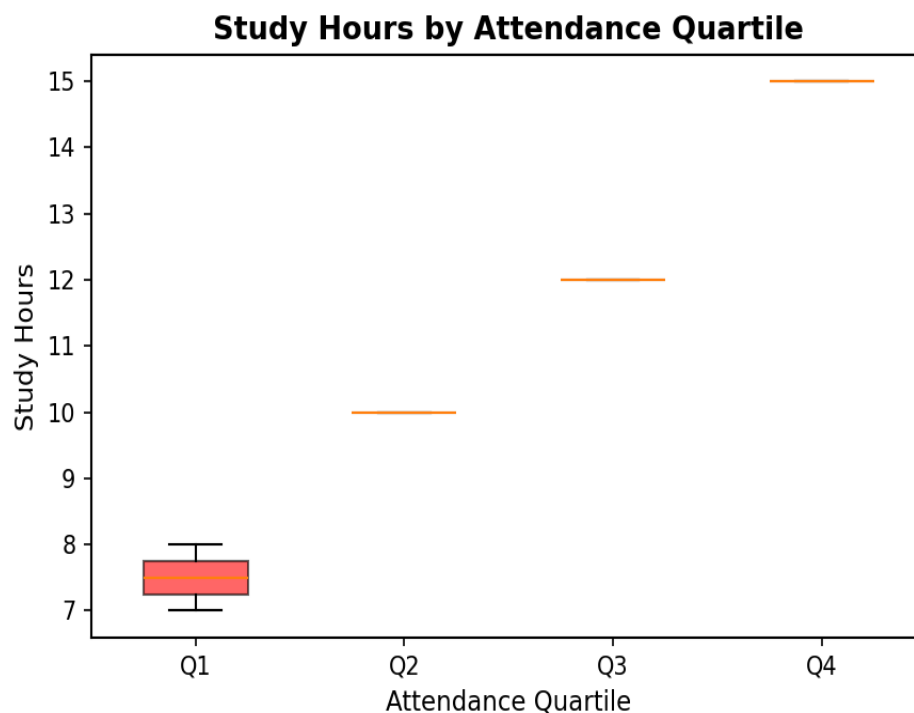
CODE:

```
# Create Attendance quartile groups
attendance_quartile <- cut(attendance,
                           breaks = quantile(attendance, probs = seq(0,1,0.25)),
                           include.lowest = TRUE,
                           labels = c("Q1","Q2","Q3","Q4"))

student_data <- data.frame(study_hours, attendance_quartile)

# Boxplot
boxplot(study_hours ~ attendance_quartile, data = student_data,
        col = c("red","green","blue","orange"),
        xlab = "Attendance Quartile",
        ylab = "Study Hours",
        main = "Study Hours by Attendance Quartile")
```

OUTPUT:



Interpretation: Students in the lower attendance quartiles tend to report fewer study hours, while those in the higher attendance quartiles show more study hours, suggesting attendance and study effort are positively related. With only five students the groups are small (often a single value per quartile), so no reliable outliers can be identified from this sample.

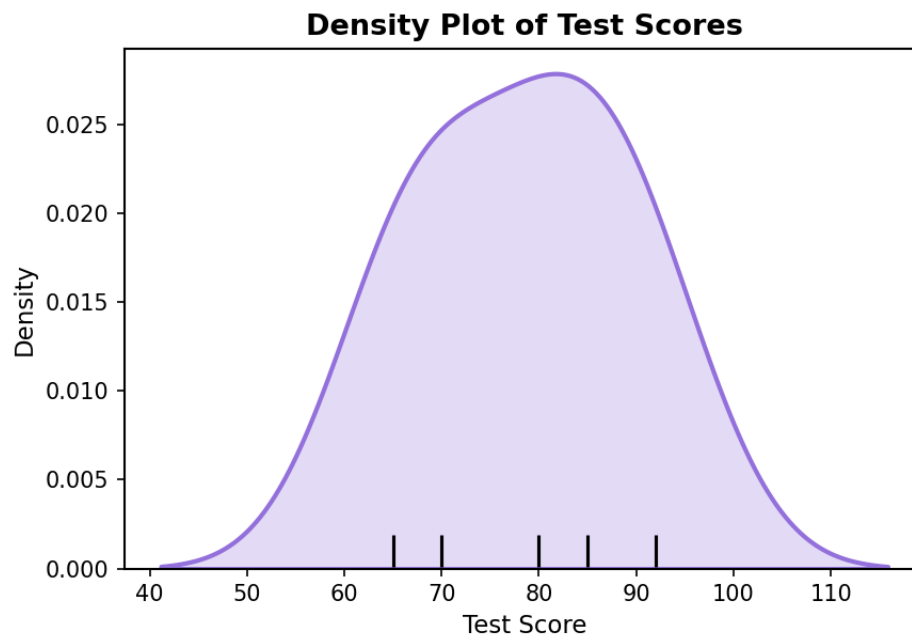
3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

CODE:

```
# Density Plot for Test_Score
density_test <- density(test_score)

plot(density_test,
     col = "mediumpurple", lwd = 2,
     xlab = "Test Score",
     ylab = "Density",
     main = "Density Plot of Test Scores")
polygon(density_test, col = rgb(0.6,0.4,0.8,0.4), border = "mediumpurple")
rug(test_score, col = "black", lwd = 2)
```

OUTPUT:



Interpretation: The density plot shows a fairly broad, single-peaked distribution of test scores between roughly 65 and 92, with no isolated points far from the rest of the data - indicating no clear outliers in this small sample.

Set 30:

Mobile App Usage Analysis

Dataset: Mobile_App_Usage.csv

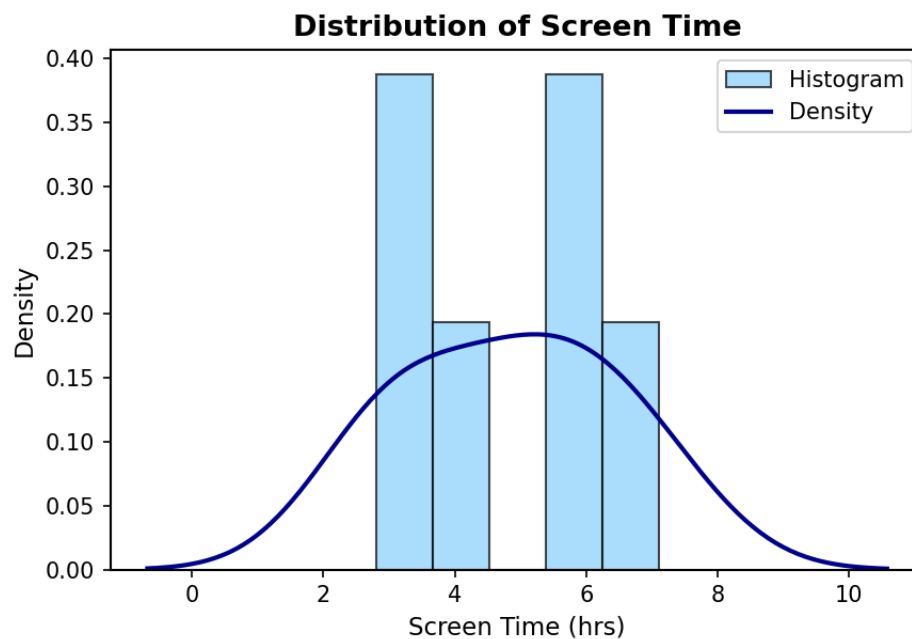
1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

CODE:

```
# Import dataset
app_data <- read.csv("Mobile_App_Usage.csv")

# Histogram with density overlay
hist(app_data$Screen_Time, freq = FALSE,
      col = "lightskyblue", border = "black",
      xlab = "Screen Time (hrs)",
      ylab = "Density",
      main = "Distribution of Screen Time")
lines(density(app_data$Screen_Time), col = "darkblue", lwd = 2)
```

OUTPUT:



Interpretation: Screen time is spread fairly evenly between about 2.8 and 7.1 hours with no strong concentration at any single value, suggesting the app is used with widely varying intensity across this small user sample.

2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

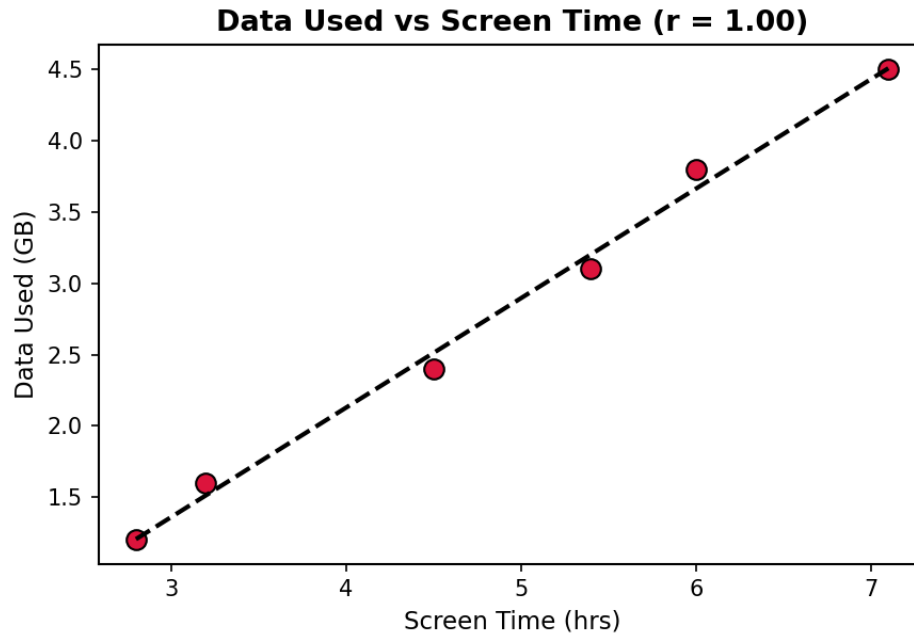
CODE:

```
# Scatter plot of Data_Used vs Screen_Time
correlation <- cor(app_data$Screen_Time, app_data$Data_Used)

plot(app_data$Screen_Time, app_data$Data_Used,
      col = "crimson", pch = 19, cex = 1.5,
      xlab = "Screen Time (hrs)",
      ylab = "Data Used (GB)",
      main = paste("Data Used vs Screen Time (r =", round(correlation, 2), ")"))
```

```
# Trend line
abline(lm(Data_Used ~ Screen_Time, data = app_data), col = "black", lwd = 2, lty = 2)
```

OUTPUT:



Interpretation: Screen time and data usage are very strongly and positively correlated - users who spend more hours on the app consistently consume more data, as shown by the tight clustering of points along the upward trend line.

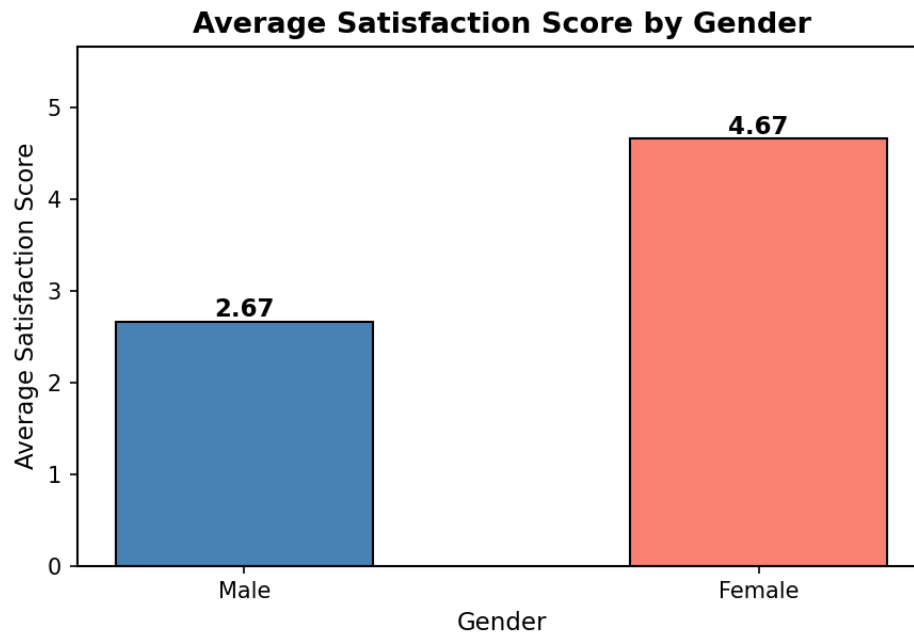
3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

CODE:

```
# Average Satisfaction by Gender
avg_satisfaction <- aggregate(Satisfaction ~ Gender, data = app_data, FUN = mean)

# Bar Chart with value labels
bar_positions <- barplot(avg_satisfaction$Satisfaction,
  names.arg = avg_satisfaction$Gender,
  col = c("salmon", "steelblue"),
  xlab = "Gender",
  ylab = "Average Satisfaction Score",
  main = "Average Satisfaction Score by Gender")
text(bar_positions, avg_satisfaction$Satisfaction + 0.1,
  labels = round(avg_satisfaction$Satisfaction, 2))
```

OUTPUT:



Interpretation: Female users report a noticeably higher average satisfaction score (about 4.7) compared to male users (about 2.7), suggesting the app experience is currently better received among female users in this sample.